A Design Case Study in the Effectiveness of Generative Models

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Abstract

Similar to the slow food movement, there may be a need to take in, and be more aware of every step throughout the design process. "Slow Design can indeed have a positive impact on wellbeing." In another context, Minsky discusses training an AI model and states "slow learning.. smoothes away uncertainties due to noise". Today with generative AI tools able to ideate and visualize incesently toward an accelerated design process, we run the risk of getting caught in a place where design happens in front of the designer more than through the designer. Minksky elaborates " it will at last become economical to match human beings in real time with really large machines. This means that we can work toward programming what will be, in effect, "thinking aids." In the years to come, we expect that these man-machine systems will share, and perhaps for a time be dominant, in our advance toward the development of "artificial intelligence."

This article considers workflows for several AI models in collaboration with designers to tackle a real world design problem. We will spend time comparing and contrasting how these models impact design workflows, making observations along the way about enhancing features and again where fidelity is lost. We will include Neural Style Transfer(NST) (an older generative AI model) which isn't as streamlined nor high fidelity. We will briefly discuss some relevent technicalities of this older model in order to illucidate opportunities of an older model with lower fidelity than the current state of the art. The lower quality, sketchy output creates ambiguity for the designer to interpret. Potentially allowing the imagination and the designers intuition to play a larger role than with some of the newer models. Working with machine learning neural style transfer (NST) can allow the designers to consider possibilities humanity may not otherwise allow ourselves to imagine. This project tests this hypothisis. Perhaps we should linger in the "man-machine system" paradigm a little longer even if we are able to go further?

Keywords

Hybrid Intelligence; Cognitive Prosthetic; Machine Learning; Schematic Design; sublime; Neural Style Transfer; Neural Networks; AI in Design; Humanity; design workflow; design exploration; early design stage;

1. Introduction

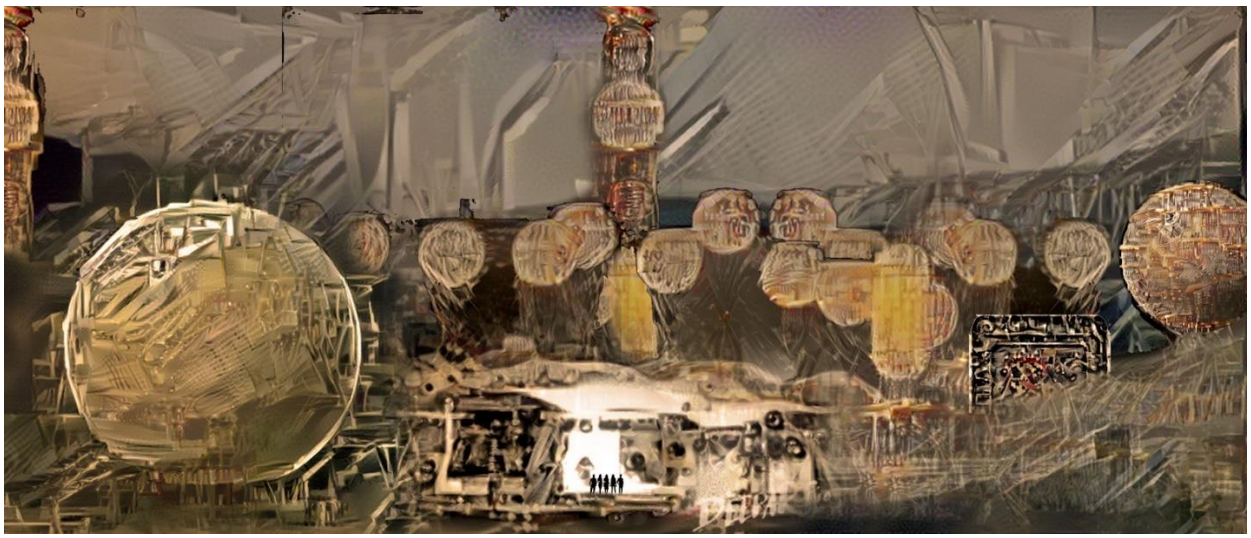


Figure 1 The work entitled “Quantum Clouds” inspired by “Instant City In a Field, Typical Set-up” by Archigram (Cook 2022). The piece was produced by the author through collaging several NST images using TensorFlow and Google CoLab, along with post processing and color calibration in photoshop. See Fig 2.3 for more detail about the NST process for this image.

This work outlines an approach to using neural style transfer (NST) in a novel way as a part of the early architectural design process. We seek to use the tool in a way unintended by its creators, for results different from many examples or use cases. (Liu et al 2021) (Liu 2021)(Wang 2021)

The work presented herein attempts to blur the lines between architectural design, story telling, digital media, computer science and other disciplines. The objective of the effort is to inspire humanity toward a brighter future through imagining and instantiating possible futures. Divergent from many other documented use cases within the design process, this use case of NST focuses on a more intuitive, iterative, feedback loop between designer and the computational process. (Fig 3) We attempt to achieve a guided vision, possibly beyond the potential conceptions of the designer in unaided circumstances to conceive of the outcomes, or potential design schemes. Built on several speculative architectural workshops, art exhibitions, and design studios at Tongji DigitalFUTURES, FutureLab Shanghai, Longxi Art Museum, and Wenzhou-Kean University, respectively, this paper seeks ontological uncertainty between the artist and a cognitive prosthetic using the machine learning technologies of a convolutional neural network (CNN), and more specifically NST in the early design phase for novel design solutions.

The collection of NST compositions represented in this article are primarily possible futures. One might ask about the merit of compositions that do not seem to be feasible or viable as constructed architecture. Yet these blue sky explorations can exist only digitally, or possibly become a part of humanity's built reality at some point in the future. **Regardless of their final instantiation, by very conception, these possibilities become exactly that, a possibility, otherwise impossible.** This paper will first outline our process. Afterward consider some of the lineage from which this work springs and the potential power of the speculative world building process. (Young 2020) Next we will consider the low hanging fruit of instantiating these places at least virtually in the metaverse and/or through VR, animation as a way to solidify their somewhat timid present reality.

2. Background

Recent advances in Artificial Intelligence have demonstrated the ability of Deep Learning models such as Convolutional Neural Networks (CNNs) to generate artistic imageries by extracting and recombining the content of an image and a style. This process of applying different styles to an image content via a CNN architecture is usually referred to as Neural Style Transfer (NST) (Gatys et al., 2016).

Due to its visually plausible results, the research in NST has amassed a significant amount of attention and effort from academic and industry practitioners. Various novel approaches have been proposed to improve the original NST architecture or extend it to cope with new application settings. In addition, many successful industrial applications have been developed and begun to deliver commercial benefits. For example, in social communication, NST applications have aided people in creating artworks and sharing their art paintings on social media platforms, which can be regarded as a new form of social communication. In that regard, Prisma[1] and deeparteffects[2] have been among the digital applications that have successfully provided NST as a service for many users. Similarly, in the contribution of (Reimann et al., 2018), authors introduced "MaeSTrO," a novel mobile application that facilitates a creative and localized style transfer editing process. Beyond social communication, NST has been a precious assisted tool for painters and designers. Nowadays, NST can make it more convenient for a painter to create a digital artwork of a particular style. Moreover, NST algorithms have produced a large panel of ready-to-use fashion components in various styles. This has immensely improved the efficiency of fashion designers and stylized CAD drawings and significantly reduced manufacturing costs (Dubey et al., 2020).

NST has also been applied in the digital entertainment sector, such as in video games, via automatically stylizing a video scene into a particular artistic style in real-time. In this convention, (Chen et al.,2018) proposed CartoonGAN, a Generative Adversarial Network (GAN) based framework, to transform photos of real-world scenes into cartoon-style images with unpaired and cartoon images for training. Another work by (Cao et al., 2018) proposed CariGANs, an unpaired GAN architecture for face to caricatures translation; the model explicitly represents the geometric exaggeration and appearance stylization.

[1] https://prisma-ai.com/
[2] https://www.deeparteffects.com/

Technically speaking, generating images in NST methods is not unique. It rather depends on the usage of the optimization process and its workflow within the NST architecture. With that, image generation in NST can be classified into 1) optimization-based methods (Li, C. et al., 2016), where the synthesized output is directly generated by optimizing the content and style loss functions within the feature space of the CNN model. This method is relatively slow but usually leads to a higher synthesized quality, 2) feed-forward methods (Chen et al., 2017) that allow the stylized output images to be generated by a single iteration, however, the output is restrained by those pre-trained style images, and 3) arbitrary methods (Huang et al., 2017) that generate stylized outputs within a single iteration given arbitrary style images. Despite improving computation efficiency, Feed-forward and arbitrary methods often sacrifice the synthesized quality.

Based on their style modeling, NST techniques can be classified into 1) statistics-based and patch-based methods (Sheng et al., 2018). The former aims to represent the image style via some statistics. In this convention, the Gram matrix is one of the first introduced and commonly used statistical metrics for NST; it computes the correlation between the various feature maps extracted at each layer of CNN. The Gram matrix has been notably improved and extended by recent NST methods. For example, (Risser et al., 2017) found that the large difference in the means and variances of feature maps is not fully reflected in the Gram matrix; to handle this imbalance, the authors integrated the histogram and TV losses within the Gram matrix. Likewise, (Li, S. et al., 2017) integrated a Laplacian loss to preserve the detailed structures of the content image and eliminate the artifacts. Other popular statistics in NST use the mean and variance for style quantification. For instance, (Li, Y. et al., 2017) presented a feed forward NST via a direct matching of feature covariance of the extracted feature activations to a given style image. Similarly, (Huang et al., 2017) introduced a novel NST method that enables arbitrary style transfer in real-time via an adaptive instance normalization layer that aligns the mean and variance of the content features with those of the style features.

Based on Markov Random Field (MRF) and CNNs, patch-based NST methods directly extract and match the local neural patches of the content and style images (Li, C. et al., 2016). In this concept, for each patch in the synthesized feature activation, a similar patch in the style activation is matched, and the style loss is calculated as the cumulative distances between those patches. Interestingly, patch and statistics-based NST methods have been combined as in the work of (Sheng et al., 2018) by firstly applying a whitening and coloring transformation to obtain aligned feature activations, on which a patch swapping transformation is then performed. Attentional-based NST has recently been proposed, which can also be regarded as patch-based style transfer. In this regard, (Park et al., 2019) and (Deng et al., 2020) introduced a style-attentional network to match the style features into the content features and a self-attention mechanism for multi-stroke style transfer respectively.

As far as the evaluation of NST algorithms is considered, the process is not entirely straightforward and remains an open and crucial challenge. Essentially, two major evaluation methodologies can be employed in the field of NST: qualitative and quantitative (Wang Z. et al., 2021). The former relies on the aesthetic judgments of observers, and its results are related to many factors (e.g., profiles of participants). In contrast, the quantitative evaluation focuses on specific metrics, which include time complexity, loss variation, etc.

3. Neural Style Transfer (NST)

In this section, we detail a Neural Style Transfer (NST) model that is used explicitly for an artistic style. In this architecture, a pretrained VGG-16 model on the ImageNet dataset is employed for the extraction of features representing the semantics of an image's content and its underlying style.

1) Image Content:

In the following, we discuss the input image from a content point of view. The input image named $\hat{x}$ is convolved with the set of existing filters in each of the CNN's layers. A layer l often includes $N_l \in \mathbb{Z}^+$ distinct filters to extract the various semantics of the image's content, and N^l feature maps each of size $M_l \in \mathbb{Z}^+$ to encode the pixel values. Hence, the output of each layer l can be represented in the form of a matrix $F_l \in \mathbb{R}^{N_l * M_l}$, with F_{ij}^l denoting the activation of the filter i at the position j of layer l.

The elements of the activation matrix F_l can therefore be used to represent and subsequently extract the content of the input image $\hat{x}$. It is significant to mention that the activations in the shallow layers tend to focus on low-level pixel information, whereas the activations of deeper layers often overlook individual pixel values and instead capture high-

level semantic information such as the presence of objects in scenes and global arrangements of objects and their relative position. Figure 1 illustrates the architecture of an NST.

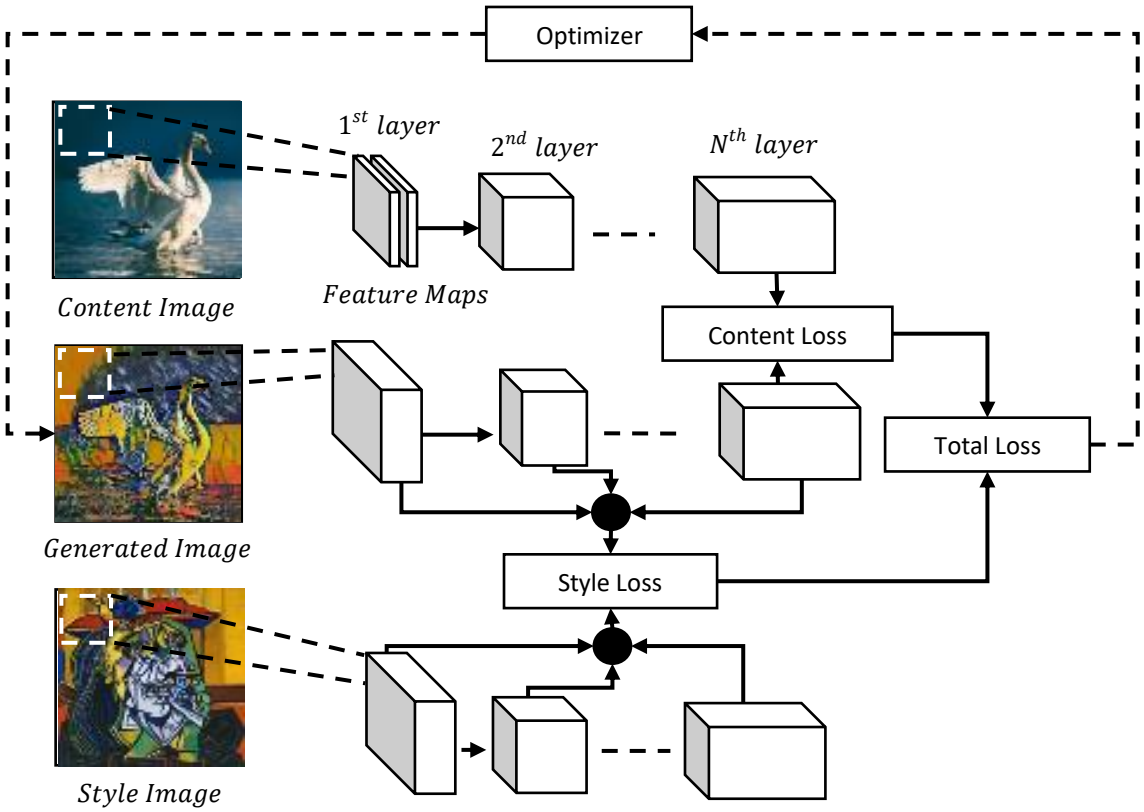


Figure 1: Illustration of the process of Neural Style Transfer

As for the content loss function, let $\hat{x}$ and $\hat{y}$ be the original and generated image, respectively, and X^l, Y^l be their respective feature maps at layer l. The contribution $C_l(\hat{x}, \hat{y})$ of each layer l to the total content loss can then be written as follows:

$$C_l(\hat{x}, \hat{g}) = \frac{1}{2}\sum_{i,j}(X_{i,j}^l - Y_{i,j}^l)$$

And the total content loss $L_{content}(\hat{x}, \hat{y})$ between $\hat{x}$ and $\hat{y}$ is

$$L_{content}(\hat{x}, \hat{y}) = \frac{1}{2}\sum_{l=0}^{L}\gamma_l C_l(\hat{x}, \hat{y})$$

Where γ_l is a hyperparameter orchestrating the contribution value of each layer l toward the content loss. An γ_l value equal to 0 indicates that the filter response is entirely omitted at that layer.

2) Image Style:

The style of an image is hard to define but is usually irrelevant to the image's content and its contained objects. One might therefore employ statistical analysis across all the feature maps to numerically represent an image style. To do so, the correlations between the feature maps are examined based on the Gram Matrix $G_l \in \mathbb{R}^{N_l * N_l}$, where G_{ij}^l is the inner product between the vectorized form of feature maps in channel i and j at layer l.

$$G_{ij}^l = \sum_k (F_{ik}^l F_{jk}^l)$$

Considering the correlations between feature maps of all layers, the style of an image is computed while disregarding the information related to the content. To compute the style loss, let us consider the following: Let $\hat{x}$ and $\hat{y}$ be the original and generated images, respectively, and X^l and Y^l be their style representations at layer l. The contribution $S_l(\hat{x}, \hat{y})$ of a given layer l to the total style loss is:

$$S_l(\hat{x}, \hat{y}) = \frac{1}{4N_l^2 M_l^2} \sum_{i,j} \left(X_{ij}^l - Y_{ij}^l\right)^2$$

Where $(N_l * M_l)$ is the size of the filter at layer l

Subsequently, the total style loss $L_{style}(\hat{x}, \hat{y})$ between $\hat{x}$ and $\hat{y}$ is:

$$L_{style}(\hat{x}, \hat{y}) = \sum_{l=0}^{L} \delta_l S_l(\hat{x}, \hat{y})$$

Where δ_l is a hyperparameter specifying the contribution of each layer to the total style loss. An δ_l value equal to 0 reflects a total absence of the style information that can be extracted from layer l.

3) Style Transfer Algorithm

In machine learning or deep learning architectures, the model parameters are optimized beforehand using an optimization technique such as Gradient Descent. As a result, a model with its optimized parameters is obtained whereby a new input can be processed. Oppositely, transferring the style of an image $\hat{a}$ onto another image $\hat{b}$ is accomplished in our architecture by repeatedly optimizing the pixel values of a newly constructed image, $\hat{c}$, so that the style of $\hat{c}$ matches the style of $\hat{a}$ and its content duplicates the content of $\hat{b}$. Therefore, the concept of a "model" optimized a priori is absent in our usage of NST. Instead, a separate optimization process occurs every time a new input takes place in the network.

More specifically, at the first stage of the process, a new image $\hat{c}$ is initialized to some random noise. Afterwards, a joined loss function comprising the style loss $L_{style}(\hat{c}, \hat{a})$ and content loss $L_{content}(\hat{c}, \hat{b})$ is optimized. The total loss function is defined as follows:

$$L_{total}(\hat{a}, \hat{b}, \hat{c}) = \alpha L_{content}(\hat{c}, \hat{b}) + \beta L_{style}(\hat{c}, \hat{a})$$

Where α and $\beta \in \mathbb{Z}^+$ are hyperparameters specifying the contribution weights of the content and style toward the total loss. A large α/β ratio indicates the content of the image $\hat{b}$ is emphasized more in the constructed image $\hat{c}$, while a small α/β value allocates more effort to the style of $\hat{c}$.

The first step in the algorithm is to extract the content and style feature maps. To do so, the style image $\hat{a}$ is passed through the network, and its style representation A^l is computed across all layers using the Gram Matrix. Subsequently, the content image $\hat{b}$ is passed through the network, and its content representation B^l at each layer is stored. Next, the constructed image $\hat{c}$, whose pixel values are randomly initialized, is passed through the network, and its style matrix G^l and content activations F^l are computed. At each layer involved in the style representation, the mean square errors between the two Gram Matrices G^l and A^l is computed for the sake of $L_{style}(\hat{c}, \hat{a})$ loss function. Additionally, the mean squared difference between F^l and B^l activations is computed for the sake of computing the loss $L_{content}(\hat{c}, \hat{b})$. The total loss $L_{total}(\hat{a}, \hat{b}, \hat{c})$ is then computed as a linear combination between the content $L_{content}(\hat{c}, \hat{b})$, and the $L_{style}(\hat{c}, \hat{a})$ loss. Moreover, as in a Gradient Descent optimizer, the derivative of the total loss is computed with respect to the pixel values of $\hat{c}$ and dynamically adjusted via back-propagation. The process is repeated until $\hat{c}$ matches the style features of $\hat{a}$ and its content features match $\hat{b}$.

4. Novel Methods Toward Future Possibilities

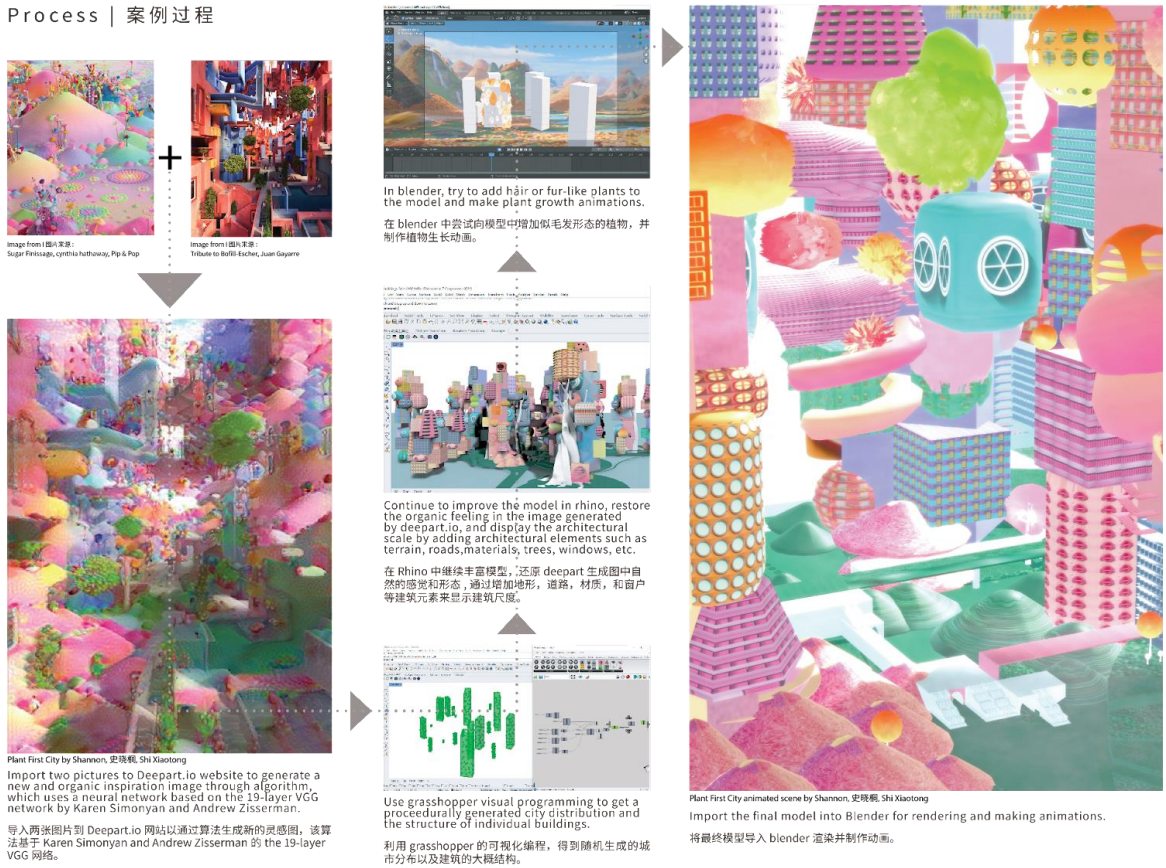


Figure 2 This diagram outlines the artistic process, of “Plant First City” by Shannon Xiaotong Shi 施晓彤. A straightforward subversion of the NST tool by combining two “non-style” images to conceive of and create a novel urban environment.

This paper reflects on a collection of works by the author and collaborators that seek to subvert the intended use case of NST. Our process utilized NST built on VGG16 and subsequently VGG19 conceived by Oxford's Visual Geometry Group (VGG, 19 denoting the number of layers in the network).(Gatys et al 2016)(Hassan 2021) The technology is designed to apply the style from, for example, a famous painting such as "Starry Night" by Van Gough, and apply it to a base image such as a photo taken with your phone.

Our unconventional approach to the process involves the intuition of the designer as a part of the style transfer and compositing process. We outline a use case for NST in the early design phase of an architectural project where the designer goes back and forth between the NST process and other inputs toward an optimized result which meets the subjective, intuitive exploratory criteria of the designer. Our process encourages the mis-use of the NST process through subverting the labeling mechanisms inherent to the NST process; encouraging a style image to be used as a base image and what may be considered a base image as a style, or in other cases we forgo the style all together and instead opt to only feed the system base images; starving the system for a style. We also encourage the compositing of multiple more traditional NST images into a single composition. After several NST exercises the design can then respond to the collected results in a collaging exercise to create outcomes that otherwise would have been impractical. (Fig. 2)

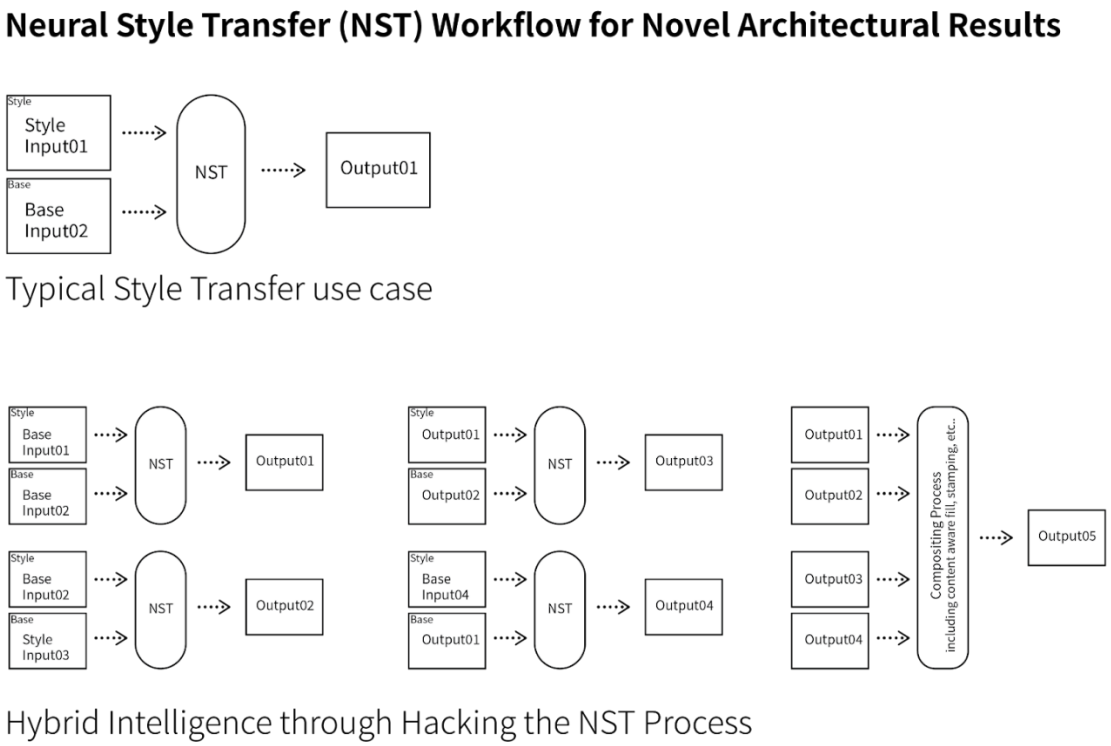


Figure 3 Comparing the process of a typical NST to the hybrid approach outlined in this paper.

The aesthetic of an NST often approaches something akin to computational impressionism.(Lee 2019) This style inherently lends itself to the potential for an emotional response from a viewer. Paired with unexpected, uncanny, or emotionally vivid elements the compositions often approach the sublime.(Kant 1914) These results can then be taken by the designer as a prompt, for further speculation and development based on a set of constraints. Our process maintains that NST output commonly provides a small peek into a potential world. It is up to the designer to take the prompt further and develop what they see in the image. After the designers develop their interpretations and speculations as a more flushed out image; often literally expanding the frame and providing elements of scale and context, a more traditional modeling process can follow, with the outcomes moving from not possible to conceived and on to a navigable experience, and possibly built. Although, there are several developing technologies with increasing capabilities able to translate single images into 3D digital models. Exploring these tools will be at the heart of future work in this research.

Several of the use cases of NST within this framework have resulted in the output weighing significantly on the final outcomes of the architectural design project. Three of the architectural design studios that have used this process have ended with results which have been directly inspired by the early visioning produced with the designer and the NST process. In the spring of 2022 the author ran an architectural design studio with co-instructor David Vardy, where NST was a core aspect to the early design phase. Later in the semester the results were curated in the “Computational Impressionism 无极之地” exhibition at the Longxi Art Museum in Zeya, Zhejiang Province.(see Fig. 5)

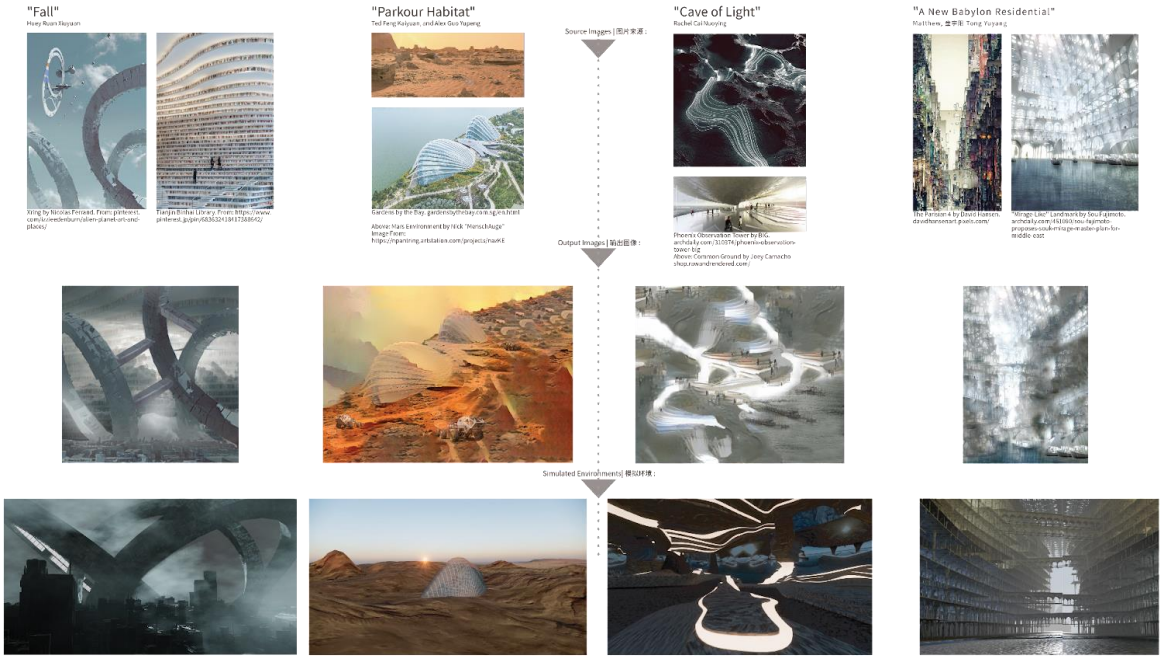


Figure 4 This figure shows other student processes examples from image selection through to virtual environment.

The students began by doing site research. collecting material samples, natural artifacts, photographs, etc. They also began processing the programmatic requirements, and in turn begin thinking about precedent projects. All of these inputs inform the subject matter input into the NST process. Yet with the distinct interest in subverting the style transfer in favor of a mixing of ideas, the outcomes demand new subjective interpretations of the original object, not allowing the observer (including the designer) to be passive about their experience viewing the outcome from the exercise. The outcomes of this superficial process generally lack scale and context. The designer is called to engage with the art. The original subject is often no longer readily recognizable. Instead, the designer must layer their own psychological imprint on the piece to interpret it. It is the opportunity of the designer to define both. As mentioned earlier this happens through an expanding of the frame to define elements such as horizon, orientation, scale, and provide some reference indicators. The results of all of these efforts are both inspiring and important in the development of hybrid intelligence within creative disciplines.



Figure 5 On the left is a photograph of the style transfer work in the Computational Impressionism 无极之地 exhibition at Longxi Art Museum. On the right is the collection of process work used to derive the "Quantum Clouds" composition.

5. Imagination: a Pioneering Effort for Humanity

Much of what drives design are performative aspects of the outcomes, yet if we are only driven by the utilitarian requirements, we will never allow ourselves the opportunity to imagine alternatives. By pushing beyond what is conceivable we enable new possibilities for the future.(Rhees 2009) This can be referred to as the development of possible futures. It is a significant part of guiding the future. Hybrid intelligent processes such as style transfer can play a key role in this and other problems humanity faces.

This research project builds on the heritage of the digital studios pioneered by Bernard Tschumi, and before him the work of Syd Mead (1933-2019), Lebbeus Woods (1940-2012), Roger Dean (1944), and others. We see the legacy of the Columbia University digital studio effort going on to create such projects as the digital Guggenheim, New City, and has even influenced Hollywood world building see examples from Alex McDowell and the World Building Media Lab at the University of Southern California. Yet this is not the only lineage of computation influencing the future of urban conceptions. We have seen, particularly in the last ten years, the influence of computational methods such as parametric, procedural, and machine learning rework the possibilities of creativity to incorporate computation as a collaborator.

Beyond those technical aspects driving creativity and innovation, there are other stirrings including more cultural ones such as the post-digital movement which has worked to flatten and colorize predominant conceptions of our future. One could argue that the digital screen is the ultimate flattening, and yet the digital screen offers so much in terms of dimensionality. The imagery produced together with the artist and the computer using NST arguably creates a new aesthetic, a new style entirely, obviously grounded in a strong foundation briefly explored above. Yet this aesthetic quite possibly may have been previously out of reach without computational collaboration.

The work that this group has undertaken is inherently optimistic and is rooted in questions such as: What kind of a world would you like to exist in if some limitation was removed? What possible futures could embody more sustainable circumstances for humanity? What is the ideal community for an individual? For industry? For community? What elements are needed in order to cultivate the most conducive environments? What methods of digital representation are optimal for expressing these ideas? How can one question traditional typologies and outdated organizational systems for the benefit of humanity? How do diverse family structures, demographics, and living/working styles affect the city?

Consider the work of Shannon Xiaotong Shi 施晓彤 with her work highlighted in Fig. 4 entitled Plant First City. In this work she considers the absurd possibility of a city where plants receive the primary and first consideration. What

would such a reality feel like? In her extra-planetary conception plants are the dominant species and in order to survive humans must live symbiotically. Even the conception of such a reality is quite intensive, let along the instantiation of it as a navigable environment, which is exactly what she was able to accomplish with her partner Luna Yuwen Wang 王宇文. (Xiaotong 2021)



Figure 6 This image was produced through a composite of NST outputs in preparation for Latika 叶心怡 Ye Xinyi’s work moving to modeling, animating and creating a navigable VR environment

Another example comes from the DigitalFutures workshop entitled “Hacking Machine Learning Style Transfer 早期设计中的机器学习工作流”(Steenblik et al 2020) The image in Fig. 6 above image by Latika can best be explained in the artist's own words:

"In the concept of Archighram’s instant city, an assembled and disassembled city is delivered… the tent is set on balloons, and balloons float above the object. Then [it all] moves on to The next location. Based on this concept, I created an instant city with a personal style. On the basis of the existing building, balloons and airships were added to create a movable building space. There is a huge energy production center in the center of the city, a few tall buildings on the edge of the city, other uses of air space are balloons and airships, and the transportation system is also shifted to the air.".

Latika developed an environment obsessed with transitory architecture, something that many have dreamed of but few have realized. Of course this fabricated virtual environment is not built, but regardless this may be the fiction that inspires the next generation to do something truly historically altering, in the same way that the pacemaker was inspired by Frankenstein. Regardless ,this effort by the artist has impacted if no one else, her future. Hopefully she will reflect on this part of her architectural training as she is considering her career path. Content creation is a growing sector with promise. Regardless, her ability to conceive of realities outside of conventional human conception has been enhanced, and her confidence entertaining computational collaborations has grown.

6. Virtual Habitation



Figure 7 This is a frame taken from “Parkour in Mars City”, an Unreal Engine simulation by Alex Guo Yupeng 郭玉鹏 and Ted Kaiyuan Feng 冯开元, based upon a series of NST compositions. (Yupeng Et al 2021)

The technologies necessary to create complex digital worlds using automated computational methods seems to still be an illusive matter. NST and related technologies by themselves fall short of being a creative force in 3D world building alone.(Zhang Et al 2020) Yet NST can be a part of a collaborative process with designers in order to directly inform and participate in “world building”(Young 2020) minimally for virtual spaces. As virtual spaces are not limited by the constraints outside of computational environments, the NST results can then be taken by the designer as a prompt, for further speculation and development based on a set of constraints.

NST seems to allow designers/architects to envision realities outside of the constraints of currently understood physics. This allows those who experience the place to engage in potentially impossible phenomena under any other circumstances. For example the work highlighted in Fig. 7 again deals with an extra-planetary parkour experience designed around the divergence in gravitational properties between our own earth and what might be; elsewhere. How does our design of space change based on this parameter? This example lets you virtually realize those differences.

These landscapes become a “digital shadow” of a physical landscape that has not yet been realized. These shadows have a life of their own outside of any physical reality. The ambition is that through these virtual prototypes we can iterate faster through possibilities toward a better future.

7. Conclusion

We have explored some pedagogical case studies, strengthening the argument that this process can yield compelling results that inspire and demand questioning, and quite possibly yield unconventional approaches to real world problems. NST has many use cases, but the most interesting may be the misuse of the tool by not simply taking the transfer at face value, but by subverting the labeling methods employed in the training set of the model to create unexpected results that take NST beyond just transferring a style toward honest unique creation. With this use case it is not difficult for the imagery to help in the process of speculating about future possibilities. Synthetic intelligence allow us to move past human conceptions considering realities we may not deem feasible. Human imagination is always a precursor to what becomes reality. Thinking beyond our current conceptions of reality is essential to move past barriers that currently hold humanity. Non-probable futures allow humanity to identify aspects worth striving for. Even if a world building exercise is not probable for instantiation within the built environment, people can inhabit these environments virtually. What future do you want? Maybe working with NST can help us to create it.

8. References

1. Bratton, Benjamin H. The Stack: On Software and Sovereignty, The MIT Press, Cambridge, MA, 2016, p. 213.
2. Cao, K., Liao, J., & Yuan, L. (2018). Carigans: Unpaired photo-to-caricature translation. arXiv preprint arXiv:1811.00222.
3. Chen, D., Yuan, L., Liao, J., Yu, N., & Hua, G. (2017). Stylebank: An explicit representation for neural image style transfer. In Proceedings of the IEEE conference on computer vision and pattern recognition (pp. 1897-1906).
4. Chen, Y., Lai, Y. K., & Liu, Y. J. (2018). Cartoongan: Generative adversarial networks for photo cartoonization. In Proceedings of the IEEE conference on computer vision and pattern recognition (pp. 9465-9474).
5. Cook, Peter. “Instant CityIn a Field, Typical Set-Up.” Archigram Archives, Archigram, archigram.net/portfolio.html.
6. Deng, Y., Tang, F., Dong, W., Sun, W., Huang, F., & Xu, C. (2020, October). Arbitrary style transfer via multi-adaptation network. In Proceedings of the 28th ACM international conference on multimedia (pp. 2719-2727).
7. Dubey, A., Bhardwaj, N., Abhinav, K., Kuriakose, S. M., Jain, S., & Arora, V. (2020). AI Assisted Apparel Design. arXiv preprint arXiv:2007.04950.
8. Gatys, L. A., A. S. Ecker and M. Bethge, "Image Style Transfer Using Convolutional Neural Networks," 2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2016, pp. 2414-2423, doi: 10.1109/CVPR.2016.265.
9. Hassan, Muneeb ul “Evolution of Style Transfer Techniques.” VGG16 - Convolutional Network for Classification and Detection, 24 Feb. 2021, neurohive.io/en/popular-networks/vgg16/.
10. Huang, X., & Belongie, S. (2017). Arbitrary style transfer in real-time with adaptive instance normalization. In Proceedings of the IEEE international conference on computer vision (pp. 1501-1510).
11. Kant, Immanuel, and J H. Bernard. Kant's Critique of Judgement. London: Macmillan, 1914.
12. Lee, Kyungho. “Computational Impressionism: Aesthetic Transference Between Impressionism and Emerging Media Arts Kyungho Lee Illinois Informatics Institute.” ISEA2019 Gwangju LUX AETERNA Proceedings of the 25th International Symposium on Electronic Art , Art Center Nabi, Seoul, Korea, pp. 668–669.
13. Li, C., & Wand, M. (2016). Combining markov random fields and convolutional neural networks for image synthesis. In Proceedings of the IEEE conference on computer vision and pattern recognition (pp. 2479-2486).
14. Li, X., Liu, S., Kautz, J., & Yang, M. H. (2019). Learning linear transformations for fast image and video style transfer. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (pp. 3809-3817).
15. Li, Y., Fang, C., Yang, J., Wang, Z., Lu, X., & Yang, M. H. (2017). Universal style transfer via feature transforms. Advances in neural information processing systems, 30.
16. Liu, C., Shen, J., Ren, Y., Zheng, H. (2021). Pipes of AI – Machine Learning Assisted 3D Modeling Design. In: Yuan, P.F., Yao, J., Yan, C., Wang, X., Leach, N. (eds) Proceedings of the 2020 DigitalFUTURES. CDRF 2020. Springer, Singapore. https://doi.org/10.1007/978-981-33-4400-6_2
17. Park, D. Y., & Lee, K. H. (2019). Arbitrary style transfer with style-attentional networks. In proceedings of the IEEE/CVF conference on computer vision and pattern recognition (pp. 5880-5888).
18. Reimann, M., Klingbeil, M., Pasewaldt, S., Semmo, A., Trapp, M., & Döllner, J. (2018, October). MaeSTrO: A mobile app for style transfer orchestration using neural networks. In 2018 International Conference on Cyberworlds (CW) (pp. 9-16). IEEE.
19. Rhees, David J. “From Frankenstein to the Pacemaker A Profile of the Bakken Museum.” IEEE Engineering in Medicine and Biology Magazine, pp. 78–79.
20. Risser, E., Wilmot, P., & Barnes, C. (2017). Stable and controllable neural texture synthesis and style transfer using histogram losses. arXiv preprint arXiv:1701.08893.

21. Shan Liu, Yun Bo, Lingling Huang, "Application of Image Style Transfer Technology in Interior Decoration Design Based on Ecological Environment", Journal of Sensors, vol. 2021, Article ID 9699110, 7 pages, 2021. https://doi.org/10.1155/2021/9699110
22. Sheng L, Lin Z, Shao J, Wang X. Avatar-net: Multi-scale zero-shot style transfer by feature decoration. In: Proceedings of the IEEE conference on computer vision and pattern recognition, 2018, p. 8242–50.
23. Steenblik, R. Spencer, Wang, Xin, Dall'Asta, Juan Carlos, Saarinen, Evan. DigitalFUTURES Tongji University, digitalfutures.world/workshops/108.html, youtube.com/playlist?list=PLBtPB9RpflEa1g1NLn70EPrxnoRVNKIBo
24. Unknown, Author, "The Amazing 5th Dimension." World Building Institute | The Future of Narrative Media, https://worldbuilding.institute/videos/the-amazing-5th-dimension.
25. Wang, Likai. (2022). Workflow for applying optimization-based design exploration to early-stage architectural design - Case study based on EvoMass. International Journal of Architectural Computing. 20. 41-60. 10.1177/14780771221082254.
26. Wang, Z., Zhao, L., Chen, H., Zuo, Z., Li, A., Xing, W., & Lu, D. (2021). Evaluate and improve the quality of neural style transfer. Computer Vision and Image Understanding, 207, 103203.
27. Xiaotong 施晓彤 Shannon Shi. "Plant First City." Shannon Xiaotong Shi 施晓彤 Personal Portfolio, Cargo Collective, xiaotongshishannon.cargo.site/.
28. Yupeng Alex 作者 Guo, and Kaiyuan Ted 冯开元 Feng. "Parkour in Mars City." Yupeng Guo Alex Portfolio, 7 June 2021.
29. Young, Liam, et al. Planet City. Australia, Uro Publications, 2020.
30. Young, Liam. "Planet City -- a Sci-Fi Vision of an Astonishing Regenerative Future." Liam Young: Planet City -- a Sci-Fi Vision of an Astonishing Regenerative Future | TED Talk, ted.com/talks/liam_young_planet_city_a_sci_fi_vision_of_an_astonishing_regenerative_future?language=en.
31. Zhang, Hang & Blasetti, Ezio. (2020). 3D Architectural Form Style Transfer through Machine Learning (Full Version). 10.13140/RG.2.2.16791.52645.
32. Zhang, Xiangyu, et al. "Accelerating very deep convolutional networks for classification and detection." IEEE transactions on pattern analysis and machine intelligence 38.10 (2015): 1943-1955.
33. "The Amazing 5th Dimension." World Building Institute | The Future of Narrative Media, worldbuilding.institute/videos/the-amazing-5th-dimension.Ruxton, C. (2016). Tea: Hydration and other health benefits. *Primary Health Care*, *26*(8), 34-42. https://doi.org/10.7748/phc.2016.e1162